

Sustainable Mobility

Cap. 4 – Alternative fuels & propulsion modes



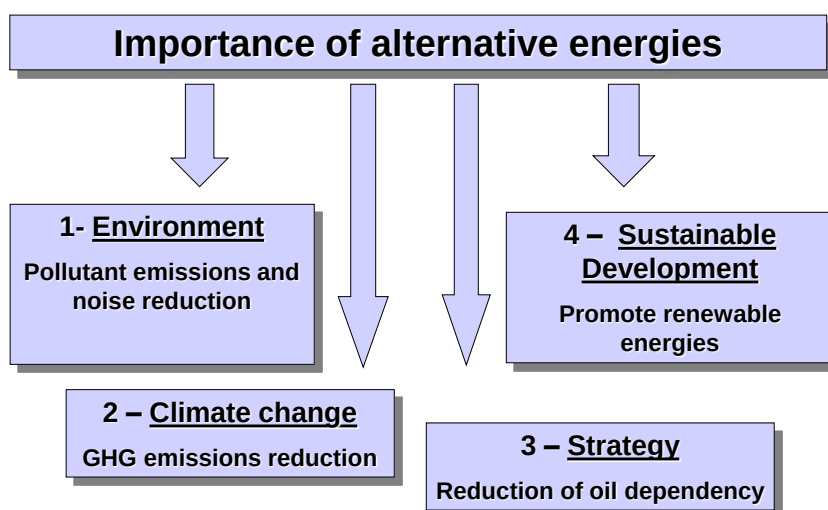
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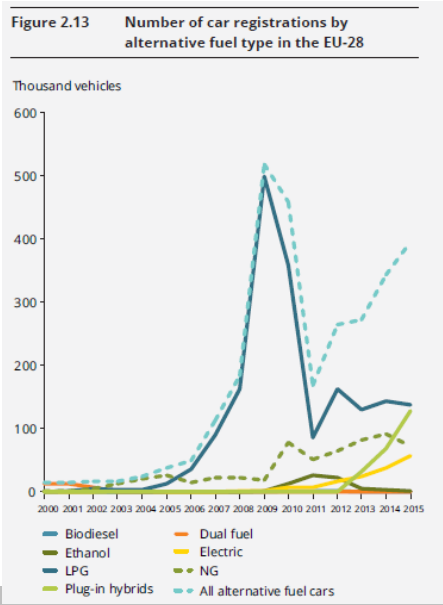
WHY THE IMPORTANCE OF ALTERNATIVE ENERGY VECTORS IN TRANSPORTATION?



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ALTERNATIVE FUELS USE



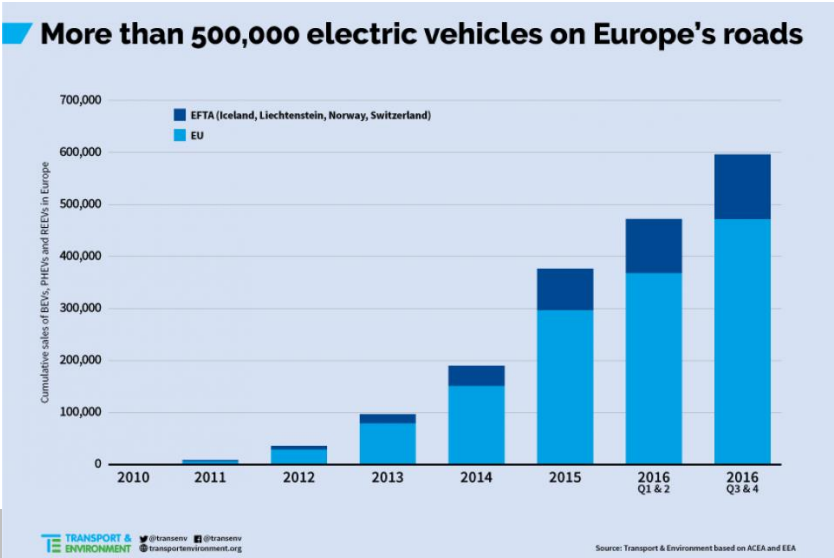
Reference: European Environment Agency (2016) TERM 2016.

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THE ACCELERATION OF EVs

Report Transport & Environment:
<https://www.transportenvironment.org/sites/te/files/Towards%20a%20European%20Market%20for%20Electro-Mobility%20report%20by%20Element%20Energy.pdf>

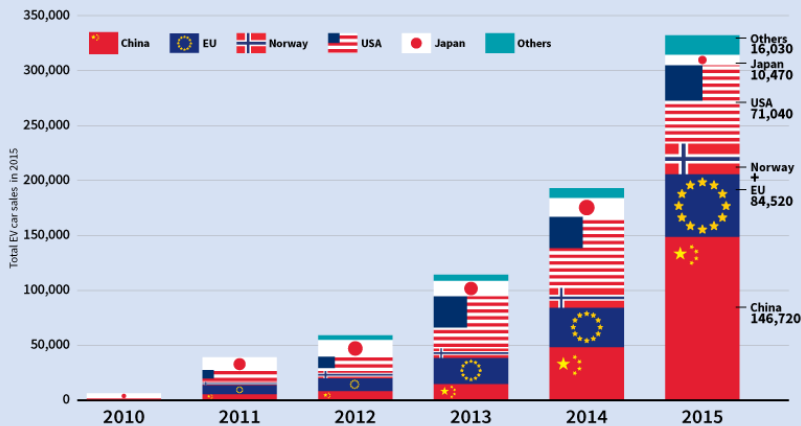


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THE ACCELERATION OF EVs

Europe: world's 2nd largest EV market



TRANSPORT & ENVIRONMENT
@transenv @transenv
transportenvironment.org

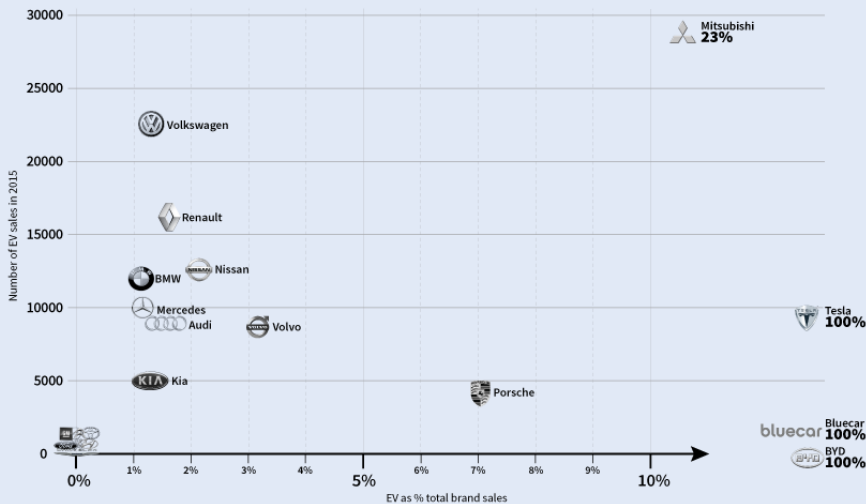
Source: adapted from IEA, Global EV outlook 2016, only BEVs

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THE ACCELERATION OF EVs

Electric vehicle manufacturers in Europe in 2015



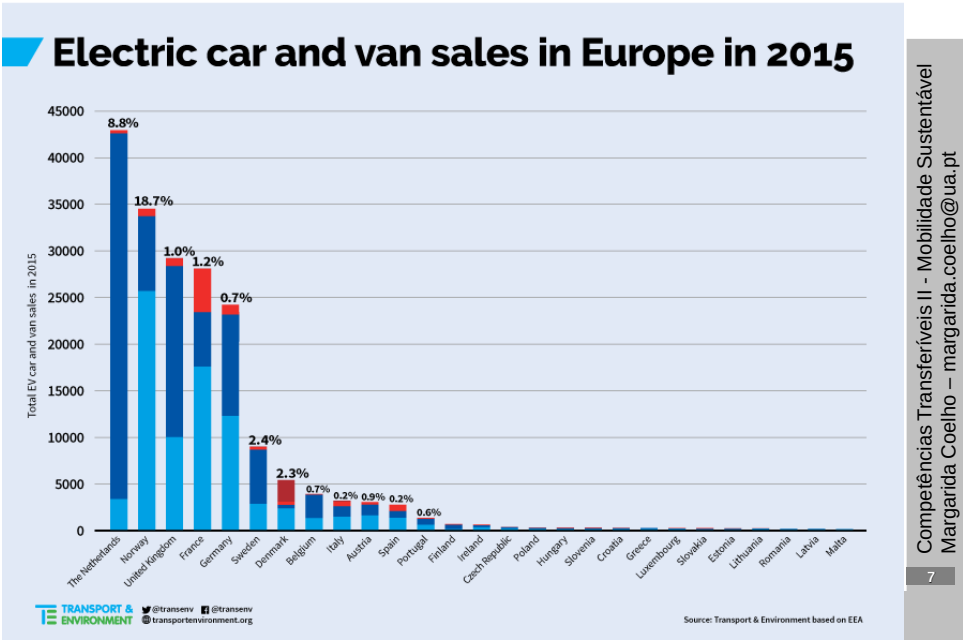
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Source: Transport & Environment based on EEA

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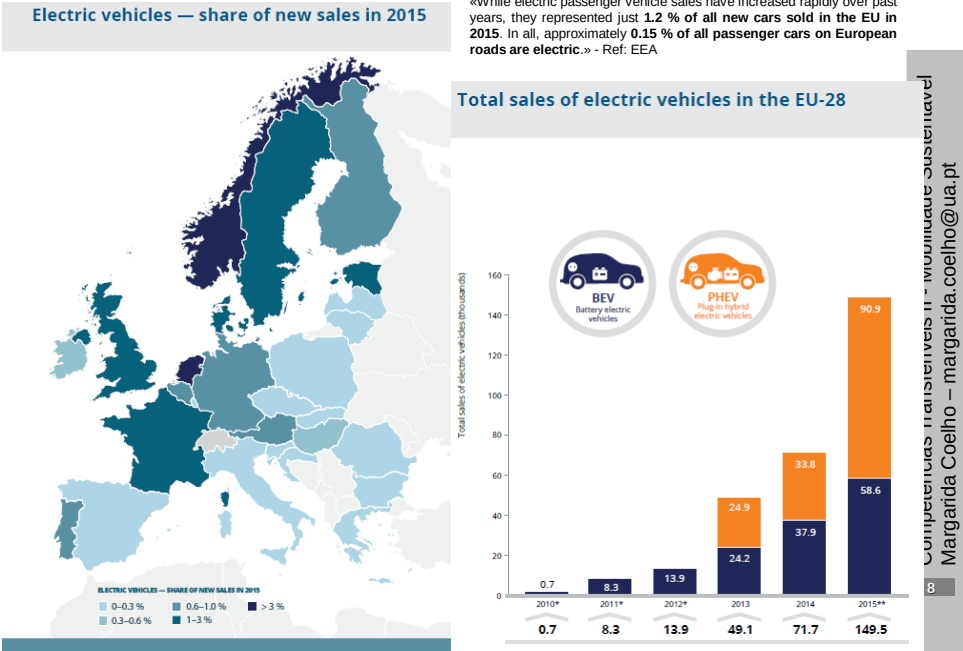
THE ACCELERATION OF EVs



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THE ACCELERATION OF EVs



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THE ACCELERATION OF EVs... in Europe!

2010: 734 EVs+hybrids
2020: 1.3 millions EVs+hybrids (54% are BEVs)

75% of the charging points are in Germany, Netherlands, Austria & Denmark

Still... per 100km of road, Portugal is the 4th country of the EU with more charging points

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THE ACCELERATION OF EVs... in Portugal!

5th country in Europe and 7th worldwide in terms of market share for EVs sales (2021)... unfortunately, now PT is not evolving at the same pace (lack of strong incentives)

In 2021, 1 in each 5 cars that were sold was an EV.
December 2021: more EVs sold (18.1% of total sold) than diesel (17.7%)
-> this never happened!

Em 2011 – 21 EVs; 671 public charging points
2021 – 130,000 EVs (includes PHEV); ~6000 public and private normal charging points + ~1000 quick, super-quick and ultra-quick charging.
2030 (predictions): 2.3 millions of Evs+PHEV

REN:
2021 – electricity consumption for electric mobility: 0.2 GWh, e.g, 0.4% of total consumption
2030 (predictions): 5.5 TWh, e.g., 10% of total consumption

Incentives:
Fundo de Apoio Ambiental: ~4000€ per EV (vs. 6000€ in Spain & Italy...)

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TARGETS

Smart and Sustainable Mobility Strategy of EC (December 2020):

By 2030:

- at least 30 million zero-emission cars will be in operation on European roads
- scheduled collective travel for journeys under 500 km should be carbon neutral
- automated mobility will be deployed at large scale
- zero-emission marine vessels will be market-ready

By 2035:

- zero-emission large aircraft will be market-ready

By 2050:

- nearly all cars, vans, buses as well as new heavy-duty vehicles will be zero-emission.

For transport to become sustainable, in practice this means:

- Boosting the uptake of zero-emission vehicles, vessels and aeroplanes, renewable & low-carbon fuels and related infrastructure - for instance by installing 3 million public charging points by 2030.
- Creating zero-emission airports and ports - for instance through new initiatives to promote sustainable aviation and maritime fuels.
- ...

Complete info at: https://transport.ec.europa.eu/transport-themes/mobility-strategy_en

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DETERMINANT FACTORS FOR THE CAR OF THE FUTURE

- **Price and scenario of oil depletion;**
- **Consumer demands;**
- **Political support.**



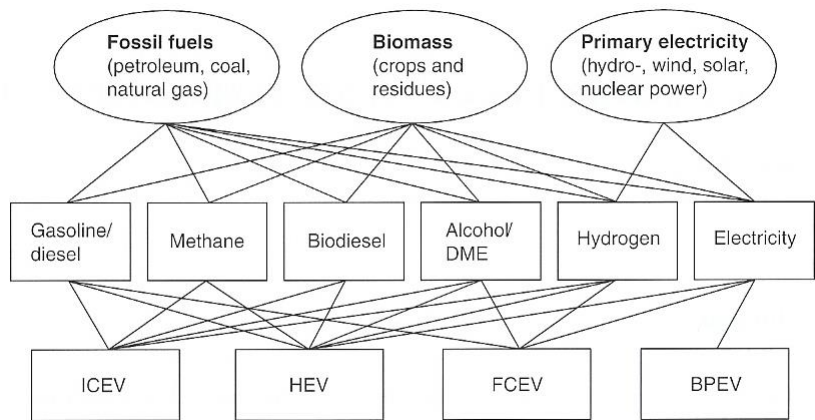
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TYPES OF ALTERNATIVE FUELS / PROPULSION MODES

From the primary energy source to final energy



Ref: Hensher, D.A. & Button, K. (eds.) Handbook of Transport and the Environment, Elsevier, 2003.

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ADVANTAGES: EMISSIONS

Vehicle technology/ reduction	CO ₂	NO _x	SO _x	PM	CO	HC
Petrol	Tailpipe: 143 g/km Europe 2010 registrations (type-approval cycle), real-world tailpipe 10–15 % higher WtW ~ 160 g/km	0.045 g/km (Euro 5 1.4–2 l)	Very low	0.017 g/km (PM ₁₀ , Euro 5 1.4–2 l)	0.594 g/km (Euro 5 1.4–2 l)	0.086 g/km (Euro 5 1.4–2 l)
Diesel	Tailpipe: 139 g/km Europe 2010 registrations (type-approval cycle), real-world tailpipe 12–20 % higher WtW ~ 143 g/km	0.433 g/km (Euro 5 < 2 l)	Very low	0.018 g/km (PM ₁₀ , Euro 5 < 2 l)	0.105 g/km (Euro 5 < 2 l)	0.016 g/km (Euro 5 < 2 l)
Hybrid electric	18–20 % less WtW than a standard petrol car (160 g CO ₂ /km WtW) or a standard diesel (143 g/km WtW) respectively	Lower than conventional cars	Lower than conventional cars	Lower than conventional cars	Lower than conventional cars	Lower than conventional cars
Mild hybrid	Similar to conventional cars	Slightly lower than conventional cars	Slightly lower than conventional cars	Slightly lower than conventional cars	Slightly lower than conventional cars	Slightly lower than conventional cars
PHEVs	WtW emissions depend on carbon intensity mix, 45 % WtW less than petrol equivalent, diesel PHEV has 5–7 % lower WtW emissions than diesel car for coal power plant production	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Very low, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation

Ref: European Environment Agency, TERM 2012.

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Vehicle technology/ reduction	CO ₂	NO _x	SO _x	PM	CO	HC
BEVs	Zero tailpipe, WTW emissions depend on carbon intensity mix, ~ 60 % WTW lower than petrol (160 g/km WTW), 445 g CO ₂ /kWh carbon intensity	Zero tailpipe, in addition to power generation	Zero tailpipe, in addition to power generation	By material wear only, in addition to power generation	Zero tailpipe, in addition to power generation	Zero tailpipe, in addition to power generation
REEV	Zero tailpipe in electric mode, similar to PHEV on average, WTW emissions depend on carbon intensity, diesel/gasoline REEVs have 34 % lower WTW GHG than diesel/gasoline cars respectively, for 467 g CO ₂ /kWh carbon intensity	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Very low, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation	Zero (urban conditions) to slightly lower than hybrids, in addition to power generation
HFCV	Zero tailpipe, WTW emissions depend on hydrogen production method: 25-100 % WTW (RES electrolysis) lower than conventional cars	Zero tailpipe, in addition to hydrogen production	Zero tailpipe, in addition to hydrogen production	By material wear only, in addition to hydrogen production	Zero tailpipe, in addition to hydrogen production	Zero tailpipe, in addition to hydrogen production
FCREV	Zero tailpipe in battery mode, WTW emissions: between a BEV and a HFCV depending on power generation	Zero tailpipe, in addition to hydrogen production/power generation	Zero tailpipe, in addition to hydrogen production/power generation	By material wear only, in addition to hydrogen production/power generation	Zero tailpipe, in addition to hydrogen production/power generation	Zero tailpipe, in addition to hydrogen production/power generation
FFV (E85)	WTW emissions depend on bioethanol production method, Tailpipe: 0-10% less than petrol	18 % lower than petrol	Zero tailpipe	34 % less than petrol	20 % lower than petrol	Similar to petrol or higher (extreme conditions)
Biodiesel (B20)	WTW emissions depend on biodiesel production method, Tailpipe: 2 % lower than Euro 3 diesel, less than 1% gain or loss depending on biodiesel source	1-2 % more than diesel	Very low, similar to conventional vehicles	10-20 % lower than diesel	5-11 % lower than diesel	10 % lower than diesel
LNG	20-24 % lower than petrol	Lower or similar to petrol	Very low, similar to conventional vehicles	Lower or similar to petrol	Lower than petrol	Similar to petrol (CH ₄ due to unburned methane)
CNG	24 % lower than petrol, retrofitted vehicles can be less efficient	Lower or similar to petrol	Very low, similar to conventional vehicles	Lower or similar to petrol	Lower than petrol	Similar to petrol (CH ₄ due to unburned methane)
LPG	10-22 % lower than petrol, retrofitted vehicles can be less efficient	Between ~ 7 % lower to 15 % higher than petrol	Very low, similar to conventional vehicles	Similar to petrol, but up to 150 % higher for retrofitted bi-fuel cars	0-57 % lower than petrol	Slightly lower than petrol

ADVANTAGES:
EMISSIONS

Legend:

- Battery Electric Vehicles (BEVs)
- Hybrid Electric Vehicles (HEVs)
- Range Extender Electric Vehicles (REEVs)
- Plug-in Hybrid Electric Vehicles (PHEVs)
- Hydrogen Fuel Cell Vehicles (HFCVs)
- Fuel-cell Range Extender vehicles (FCREVs)
- Flexi-fuel Vehicles (FFVs)
- Compressed Natural Gas (CNG)
- Liquefied Natural Gas (LNG)
- Liquefied Petroleum Gas (LPG)

Ref: **European Environment Agency, TERM 2012.**

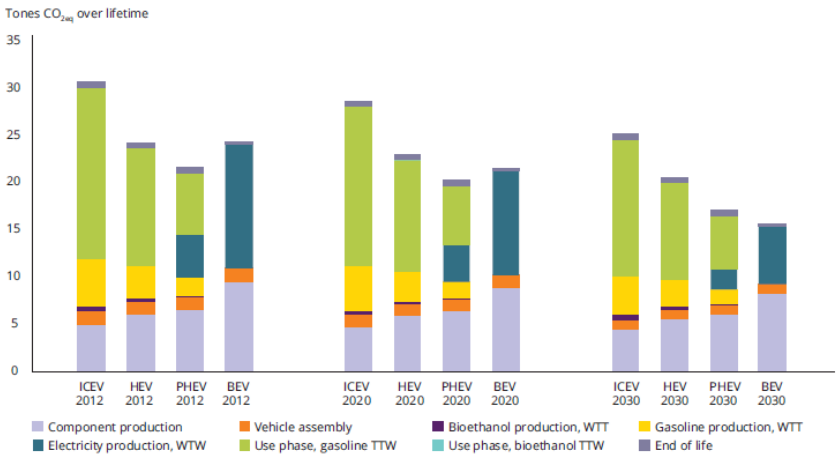
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THE IMPORTANCE OF A COMPLETE LIFE CYCLE ANALYSIS

Figure 5.2 Lifetime CO₂ emissions associated with production, fuel and energy use and disposal for medium-sized cars with different engine technologies



Note: Comparison of different medium-sized (gasoline) cars' life cycle CO₂-equivalent impacts over time (2012 as reference scenario). ICEV, internal combustion engine vehicle; HEV, hybrid electric vehicle; PHEV, plug-in hybrid electric vehicle; BEV, battery electric vehicle.

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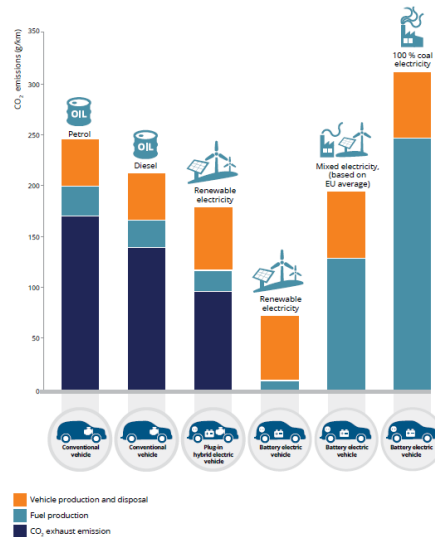


THE IMPORTANCE OF A COMPLETE LIFE CYCLE ANALYSIS

- Understanding the environmental impacts of any product requires looking at its entire **lifecycle**: from the extraction of raw materials, to the pollution generated during its manufacture, operation and disposal, through to the impacts of the residual waste remaining after disposal or recycling.
- Production of EVs is typically more energy-intensive than conventional vehicle manufacture. It needs **~70% more primary energy to make BEVs than conventional vehicles**, mainly for the electric engine systems and batteries.
- The **source of the electricity** used during its operation plays a large role in the overall environmental impacts.
- End-of-life**: the large batteries and additional electrical components available for recycling, including the electric motor and its magnets, distinguish EVs from conventional vehicles.

Ref. EEA, Electric vehicles in Europe, 2016.

Range of life-cycle CO₂ emissions for different vehicle and fuel types



Note: The values are estimated for an average mid-class vehicle, based on 220 000 km.
 Source: 2014-2015 authors' own calculations



NATURAL GAS

NATURAL GAS (~ methane CH₄)

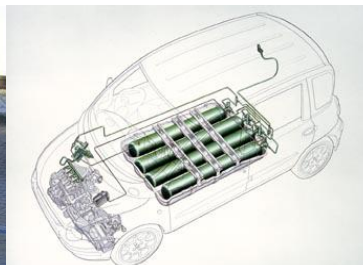
Production: natural reserves

Advantages: Low emissions (???)

Disadvantages: supply, distribution network, on-board storage

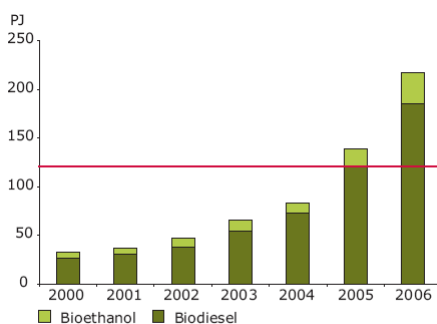
Applications: bus fleets

Potential: low (for light duty vehicles) – medium/high (buses)
 (dependent on strategic decisions)



TYPES OF BIOFUELS

- Biodiesel - fuel derived from biological sources (animals or plants).
- Bioethanol - fuel derived from biomass (sugar cane, maize, beet, sorghum, potatoes, etc.)
- Alcohol - methanol, ethanol, propanol and butanol (produced biologically from the action of bacteria)



Source: EurObservER.

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TYPES OF BIOFUELS

1st generation biofuels:

Bioethanol

Biodiesel

Bio-ETBE (ethyl tert-butyl ether)

Biogas

Pure vegetable oils

Animal fats

Recycled food oils from restaurants, domestic sector and food industry

2nd generation biofuels:

Bioethanol cellulosic

Bio-methanol

Bio-MTBE

DME - Bio-dimethyl ether

Synthetic biofuels

Microalgal biofuels

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BIODIESEL

Production: vegetable oil (sunflower, peanut,) + % diesel

Advantages: low emissions (?); renewable energy; biodegradable; non toxic; combustion has lower emissions than gas oil; works on existing diesel engines; lubrication index higher than diesel

Disadvantages: motor efficiency (5-10%); its production is more expensive than that of diesel fuel (at least for now! ...); degrades some plastic accessories (used in older cars)

Applications: agricultural fleets, cargo and passenger transport vehicles

Potentials: averages (type of production, quantities and advantages)



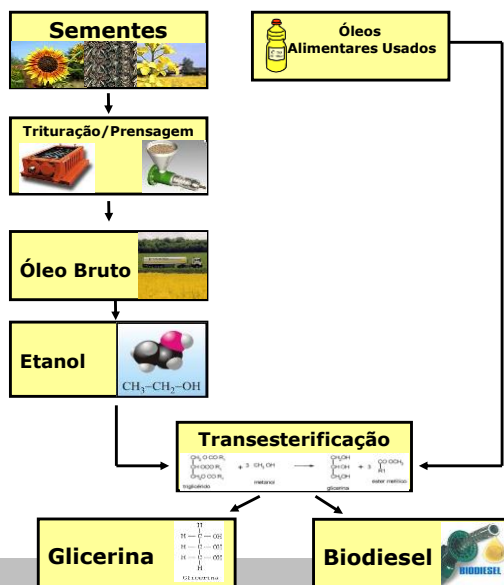
BIODIESEL CYCLE

➤ Biodiesel life cycle

- Phases:
 - Storage
 - Fuel;
 - Use in the vehicle;

➤ How to obtain biodiesel:

- Extraction;
- Transesterification;



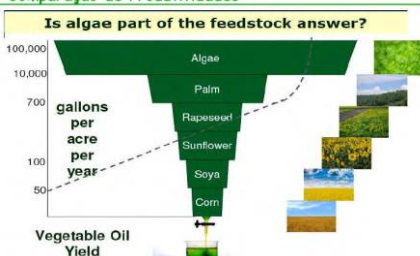
MICROALGAL BIODIESEL

It is presented as the ideal solution at a time when biofuels are responsible for the food crisis.

Benefits:

- Can be used on land unsuitable for conventional farming
- They may use salt, brackish and residual water of marginal quality
- Consume CO₂ and produce oxygen
- They can use exhaust gases from industrial plants (6-8%) or gases produced in aerobic fermentations
- Productivity much higher than traditional oleaginous (> 30 000 L / ha / year) (>> palm)
- They can reach CO₂ removal rates:> 2 kg CO₂ / kg Microalgal biomass

Comparação de Produtividades



Ref: Fernanda Rosa (2009) *Biocombustíveis: Uma alternativa no presente. Será uma alternativa no futuro?*. Ordem dos Engenheiros.

MICROALGAL BIODIESEL



Exemplos de fotobioreactores para microalgas



Research made in DEM: PROJECT ENERBIOALGAE



Partners: University of Vigo (Coordinator); INEGA;
University of Aveiro; University of Pau; University of Almeria



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ETHANOL (C_2H_5OH)

Production: fermentation of sugar cane, corn, beet, grapes ...

Advantages: low emissions, high I.O. (113, superior to gasoline
⇒ better thermal efficiency), can be used in gasoline engines
alone or mixed with gasoline (bi-fuel vehicles)

Disadvantages: 70% of autonomy, price, more expensive
processing, degrades some plastic accessories

Applications: oxygenation of gasolines (bio-ethanol; gasohol);
alternative fuel in Brazil

Potentials: average (type of production, quantities))

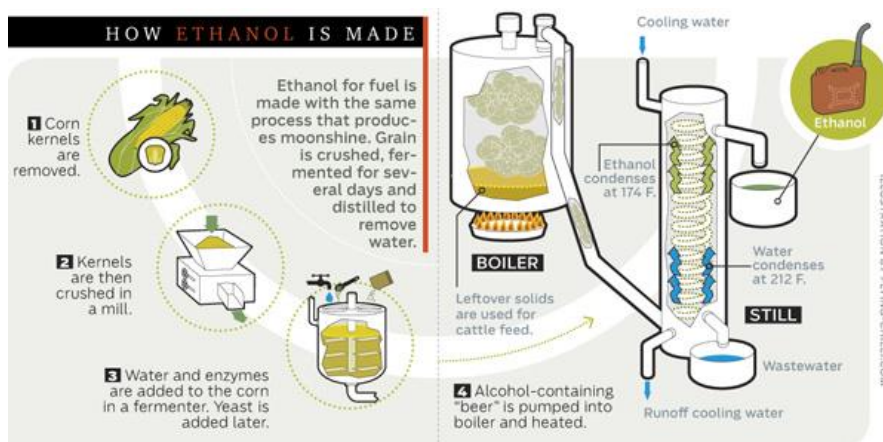


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PRODUCTION OF ETHANOL (C_2H_5OH)

From corn



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CORN ETHANOL

-Almost all US ethanol is distilled from corn

- Most ethanol is sold as a gasoline additive or as E85 (85% ethanol, 15% gasoline)

- Production in the USA: 18.3 billion liters

- Energy balance: fossil fuel energy used to produce fuel (input) compared to fuel energy (output) → 1: 1.3 or less (input: output)

-GHG emissions (production and use): 2.44 kg / l gasoline vs. 1.94 kg / l corn ethanol (21% lower)

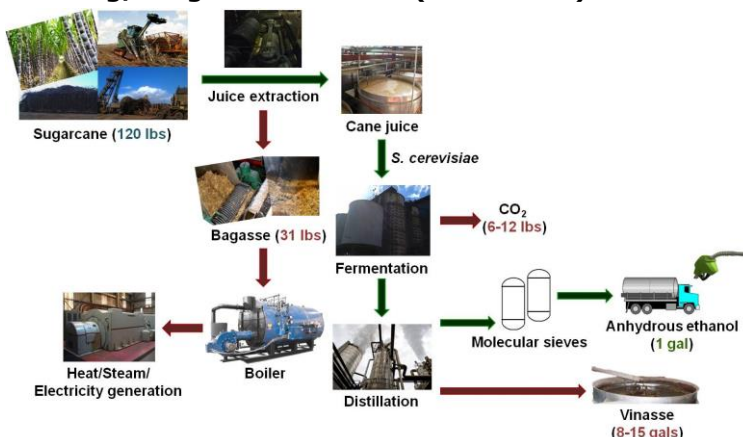
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Ref: US DOE, US EPA

SUGAR CANE ETHANOL

- Brazil: sugarcane yields 5700-7600 l/hectare
- Energy balance: fossil fuel energy used to produce the fuel (input) compared to fuel energy (output) → 1 : 8 (input:output)
- GHG emissions (production and use): 2.44 kg/l gasoline vs. 1.08 kg/l sugarcane ethanol (56% lower)



CELLULOSIC ETHANOL

- Perennial grasses can replace up to 13% of the world's oil consumption (if an efficient method is developed to transform the cellulosic matter of the plants into ethanol)
- Cellulosic ethanol sources: agricultural, forest and municipal solid waste; paper folder; fast growing vegetation
- Production: still under development in the USA, Canada, Sweden, Denmark and Spain; there is no industrial production
- Energy balance: energy in fossil fuel used to produce fuel (input) compared to fuel energy (output) → 1: 2-36 (depending on the method of production)
- GHG emissions (production and use): 2.44 kg / l gasoline vs. 0.23 kg / l sugarcane ethanol (91% lower)

FUTURE IN BIOETHANOL

Via bioquímica para biocombustíveis 2ª geração - Bioetanol de 2ª geração

1. Deixar de utilizar 7-10% da cultura cerealífera

Grão de amido (glucose)



Para utilizar 100% da planta

Grão de amido (glucose)

Celulose (glucose)

Hemiceluloses (pentoses e hexoses)

2. Resíduos agrícolas e florestais



Processos tecnológicos mais complexos

Custo de processamento mais elevado

Conceito de bio-refinaria

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Ref: Fernanda Rosa (2009) *Biocombustíveis: Uma alternativa no presente. Será uma alternativa no futuro?*. Ordem dos Engenheiros.

BIOFUELS: MAJOR CONCERNS

- Competition with food crops for the same soils
- Risk of distortion of the food market
- Effective reductions in GHG emissions (Life Cycle Analysis)
- NOx Emissions
- Real energy gains (what cultures?)
- Danger of deforestation and loss of biodiversity
- Need for environmental sustainability criteria for the production of biofuels

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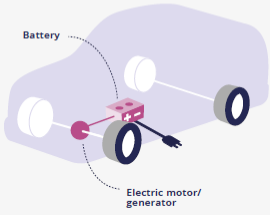
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HYBRID AND ELECTRIC VEHICLES

EV TYPES

Battery electric vehicle

Battery electric vehicles are powered by an electric motor and battery with plug-in charging.



ADVANTAGES

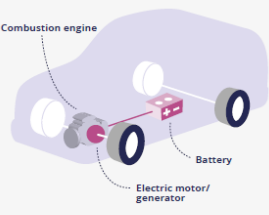
- HIGHER EFFICIENCY
- HOME/WORKPLACE RECHARGE
- LOW ENGINE NOISE
- ZERO EXHAUST EMISSIONS

DISADVANTAGES

- FEWER RECHARGING STATIONS
- LONG TIME TO RECHARGE
- SHORT DRIVING RANGE

Hybrid electric vehicle

Hybrid electric vehicles combine a conventional (petrol/diesel) engine and a small electric motor/battery charged via regenerative braking or the engine.



ADVANTAGES

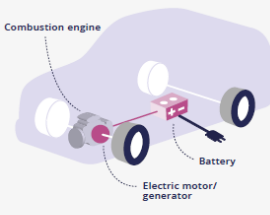
- HIGHER EFFICIENCY
- MANY REFUELLING STATIONS

DISADVANTAGES

- EXHAUST EMISSIONS
- FOSSIL FUEL DEPENDENCY
- HIGHER ENGINE NOISE
- TECHNOLOGICAL COMPLEXITY

Plug-in hybrid electric vehicle

Plug-in hybrid electric vehicles have a conventional (petrol/diesel) engine complemented with an electric motor/battery with plug-in charging.



ADVANTAGES

- HIGHER EFFICIENCY
- HOME/WORKPLACE RECHARGE
- MANY REFUELLING STATIONS

DISADVANTAGES

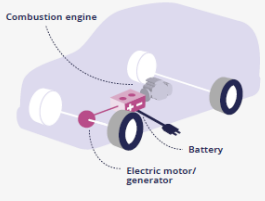
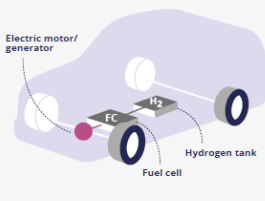
- TECHNOLOGICAL COMPLEXITY

Indicative electric driving range:

80–400 km.	0–10km	20–85 km
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Ref. EEA, Electric vehicles in Europe, 2016.

EV TYPES

Range-extended electric vehicle	Fuel cell electric vehicle
Range extended electric vehicles are powered by an electric motor and plug-in battery, with an auxiliary combustion engine used only to supplement battery charging. 	Fuel cell electric vehicles use a fuel cell to create on-board electricity, generally using compressed hydrogen and oxygen from the air. 
ADVANTAGES - HIGHER EFFICIENCY - HOME/WORKPLACE RECHARGE - MANY RECHARGING STATIONS	ADVANTAGES - HIGHER EFFICIENCY - LOW ENGINE NOISE - ZERO EXHAUST EMISSIONS
DISADVANTAGES TECHNOLOGICAL COMPLEXITY	DISADVANTAGES - LIMITED COMMERCIAL AVAILABILITY - LACKING RECHARGING STATIONS - TECHNOLOGICAL COMPLEXITY
Indicative electric driving range: 70-145 km	160-500 km

Ref. EEA, Electric vehicles in Europe, 2016.

HYBRID TECHNOLOGY

A hybrid vehicle is a vehicle that uses two energy sources to move.

In this section: hybrid-electric.

Some interesting advantages, coming from the electric origin of the movement:

- Regenerative braking (minimizes energy lost during braking);
- Decreased fuel consumption;
- Decrease in emissions (thermal engine works in more efficient regimes and consumes less fuel).

HYBRID / ELECTRIC VEHICLES

Production: electricity (electric); electricity + gasoline or diesel (hybrids)

Advantages: electric motor $\Rightarrow$ zero LOCAL emissions

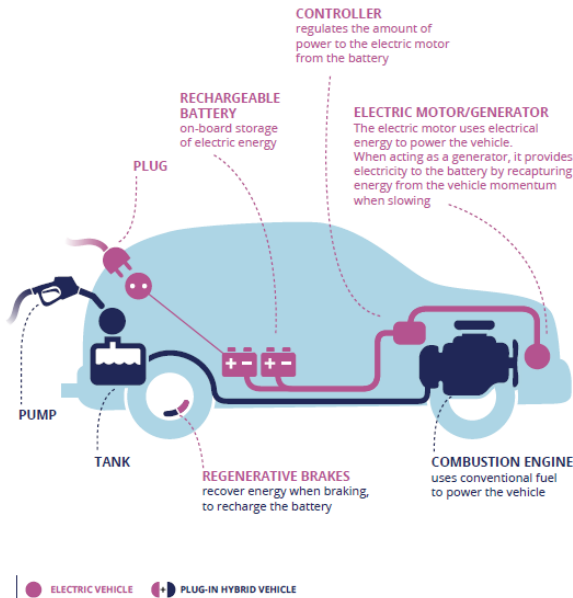
Disadvantages: recharging (electric vehicles); battery

Applications: fleets, private cars

Potential: Very High



MAIN COMPONENTS OF EVs



Ref. EEA, Electric vehicles
in Europe, 2016.



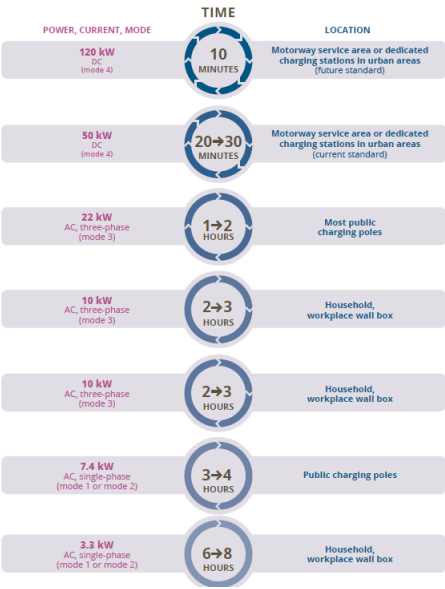
• **Plug-in charging**

It is used by the vast majority of current BEVs and PHEVs in Europe. Vehicles are physically connected to a charging point using a cable and a plug. Plug-in charging can occur wherever charging stations are located: at homes, in public streets or on commercial or private premises. EVs can be charged using normal household sockets, but this is slow because normal domestic sockets provide only a low amount of electric current. It can take ~8h for a typical charge. This can be quite suitable for overnight charging. Faster plug-in charging requires specialised infrastructure. Most public plug-in stations established at a city, regional or national level offer only normal-speed charging.

Ref. EEA, Electric vehicles in Europe, 2016.

3 CHARGING TYPES

Charging times to provide 100 km of battery electric vehicle driving



• **Battery swapping**

It involves replacing a used battery with a fully charged one at a special swapping station.

At present, no major providers in Europe offer battery swapping. A number of barriers have prevented battery-swapping technology from becoming widespread, including the lack of EV models that support battery swapping, no standard type or size of battery, and the high cost of developing the associated charging and swapping infrastructure.

• **Wireless charging**

Also known as induction charging, does not require a fixed physical connection between the charging facility and the vehicle. Instead, the system creates a localised electromagnetic field around a charging pad, which is activated when an electric vehicle with a corresponding pad is positioned above it.

The wireless method currently operates at only a selected few pilot locations and is yet to be used commercially.

Examples of inductive charging pilot projects include wireless charging for buses at bus stations in Belgium, Germany, the Netherlands and the United Kingdom, as well as some pilot testing for users of electric cars in Sweden.

3 CHARGING TYPES

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Honda



BMW i3



Renault

Electric buses

They have great potential for future use in cities.

Electric buses using high-capacity batteries are currently in use or under testing in a number of cities. There are > 500 battery electric buses in the EU-28. The highest numbers are in Austria, Belgium and the Netherlands.

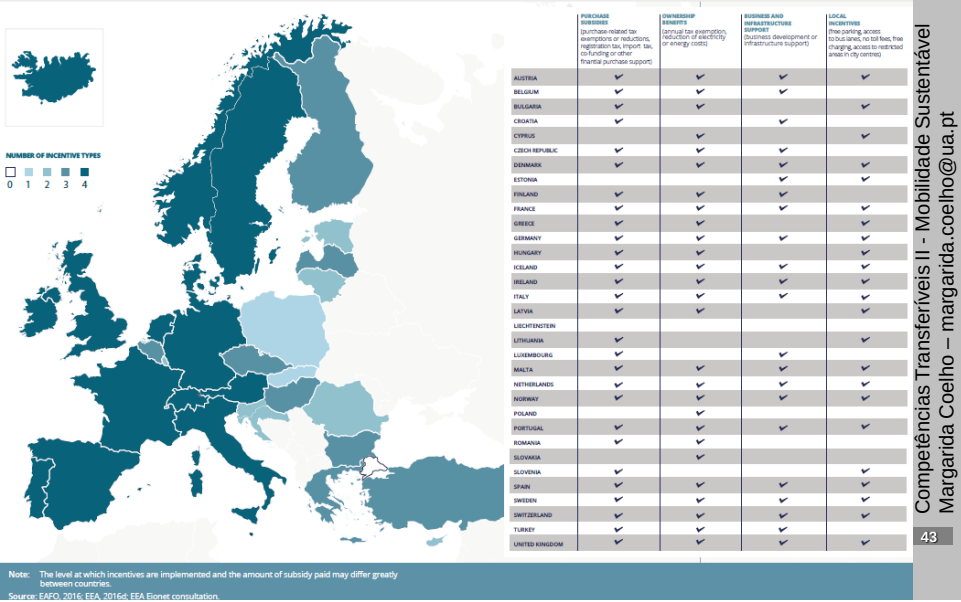
Plug-in hybrid electric buses offer more flexibility than battery electric buses and are increasingly being used in many cities across Europe including Budapest, Copenhagen, London, Oslo and Warsaw. Both battery electric and hybrid buses are more expensive than conventional buses, and can also suffer from temperature-sensitive energy consumption. Certain electric bus models can be charged overnight, while others, with smaller batteries, must also be charged during the day.



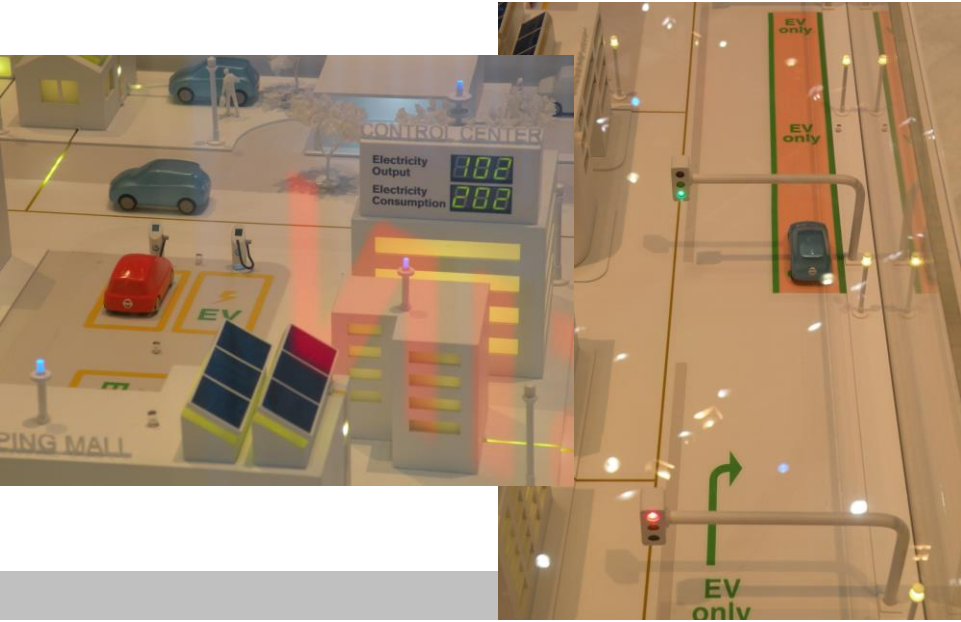
©MCCoelho

THE IMPORTANCE OF INCENTIVES

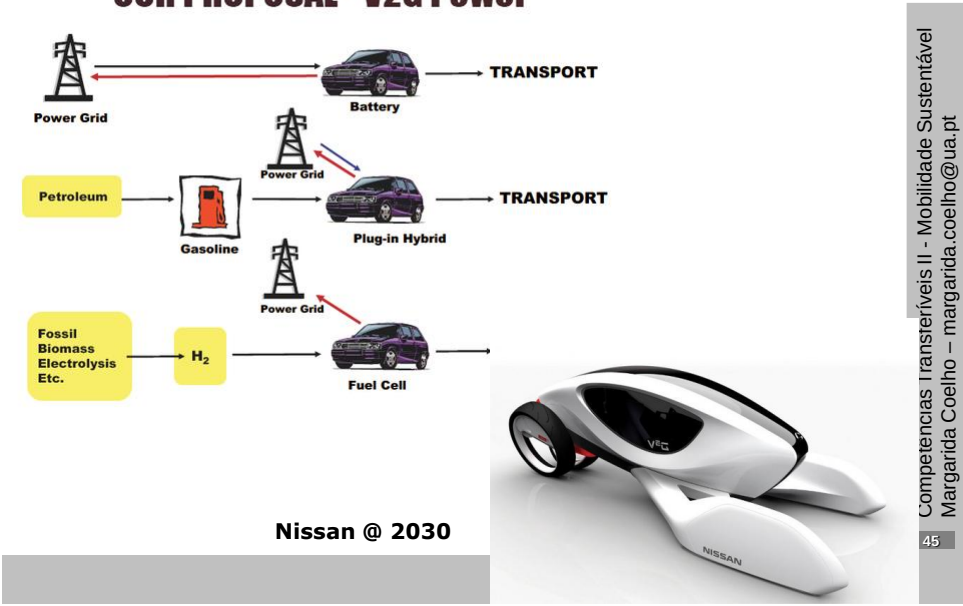
Use of incentives for electric cars across Europe



THE INFRASTRUCTURE OF THE FUTURE FOR EVs



OUR PROPOSAL - V2G Power



HYDROGEN AS AN ENERGY VECTOR

Production: electrolysis of water, natural gas, oil...

Advantages: low emissions (H₂O), high IO, compatible with fuel cells

Disadvantages: cost, production, transportation, supply, storage, security,

Applications: fuel cells, internal combustion engines

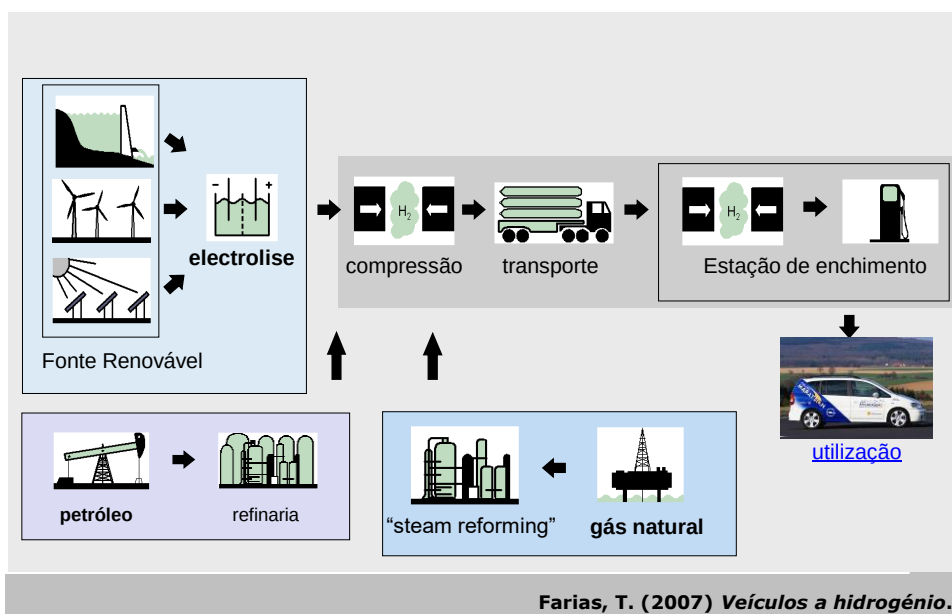
Potentialities: It all depends on the future of fuel cells



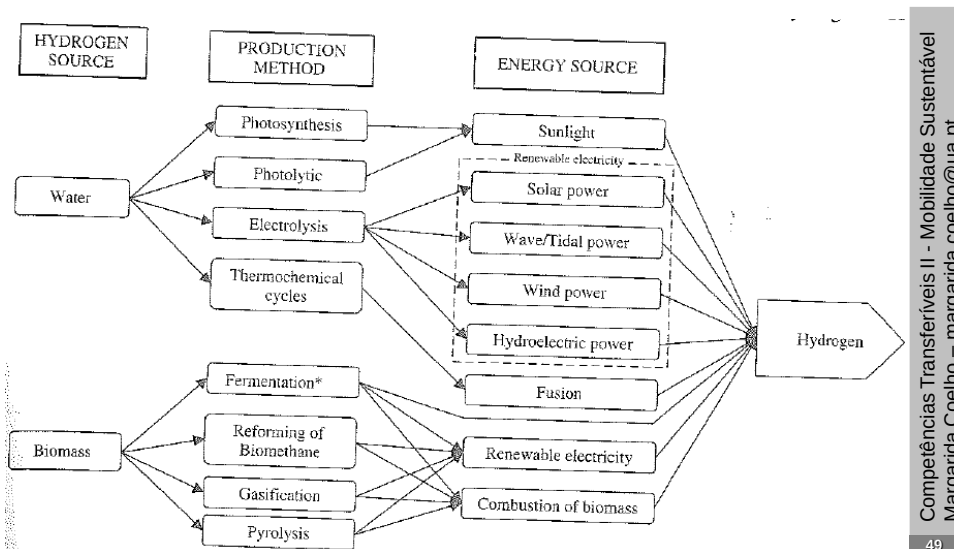
<http://auto.howstuffworks.com/fuel-cell.htm>

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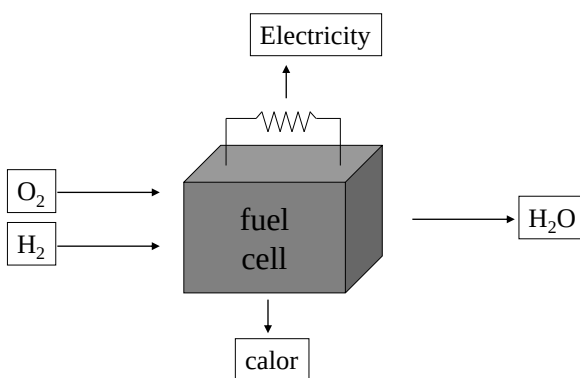
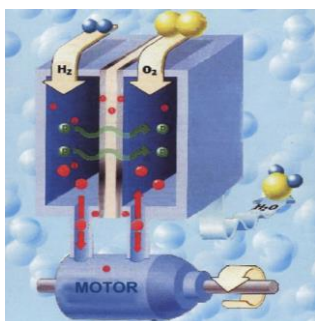
HOW TO PRODUCE H₂?



Fonte: Environmental Conscious Transportation, 2008.

FUEL CELLS

They are energy devices by electrochemical conversion
(reversal of electrolysis)





SAFETY OF AN H₂ VEHICLE

- Benefits:
H₂ is lighter than air and disperses easily
H₂ has lower radiant heat ⇒ <secondary fire risk

- "Disadvantages":
H₂ is odorless and colorless → installation of sensors in vehicles
Combustion characteristics (see Table) → for low concentrations (<10%), the energy required for the ignition of H₂ is high (~ the energy required for the ignition of NG and gasoline, if the H₂ conc. Approximates stoichiometric mixture of 29% H₂ in air), the ignition energy ↓ to 1/15 that required for ignition of GN (or 1/10 of gasoline).

	Hydrogen	Gasoline Vapor	Natural Gas
Flammability Limits (in air)	4-74%	1.4-7.6%	5.3-15%
Explosion Limits (in air)	18.3-59.0%	1.1-3.3%	5.7-14%
Ignition Energy (mJ)	0.02	0.20	0.29
Flame Temp. in air (°C)	2045	2197	1875
Stoichiometric Mixture (most easily ignited in air)	29%	2%	9%

Fonte: www.eere.energy.gov/hydrogenandfuelcells/

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SAFETY OF AN H₂ VEHICLE

Ex. Honda:
«H₂ Safety: Sensors are located throughout the vehicle to provide a warning in the unlikely event of a H₂ leak. Should such a leak occur, the ventilation system is activated and an automatic system closes the main cut-off valves on the H₂ tank or supply lines as necessary. The high-voltage lines are electrically isolated. In the event of a collision, the system controller automatically shuts off the flow of H₂ and electric current. ».
«Refueling Safety: To prevent reverse flow from the tank, the H₂ filler inlet has an integrated check valve. The fuel intake mechanism is also designed to prevent contamination by other gases or the connection of nozzles designed for H₂ stored at incompatible pressure levels.

Hydrogen Safety

The FCX Clarity Fuel Cell Electric Vehicle (FCEV) is not an experimental or prototype vehicle. It is a full-featured sedan that can be driven with both pride and confidence.

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Ref: <http://automobiles.honda.com/fcx-clarity/hydrogen-safety.aspx>

EX. FUEL CELL VEHICLES

Honda FCX-Clarity



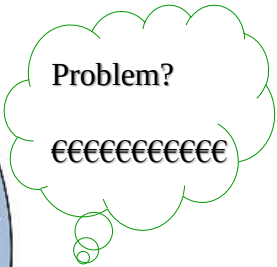
Renault Scenic ZEV H2



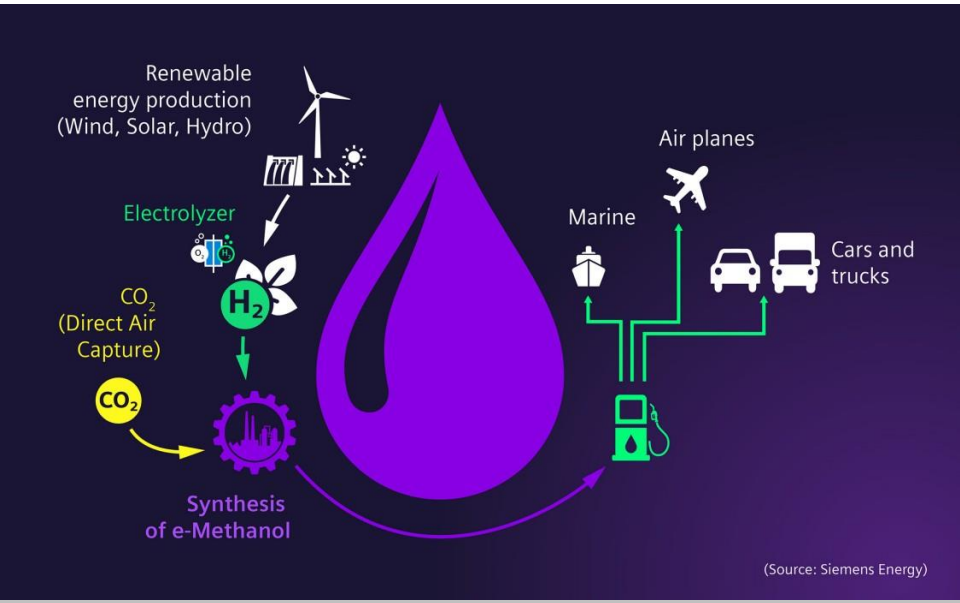
Toyota Mirai



H2 ECONOMY



ENERGY FOR TRANSPORT

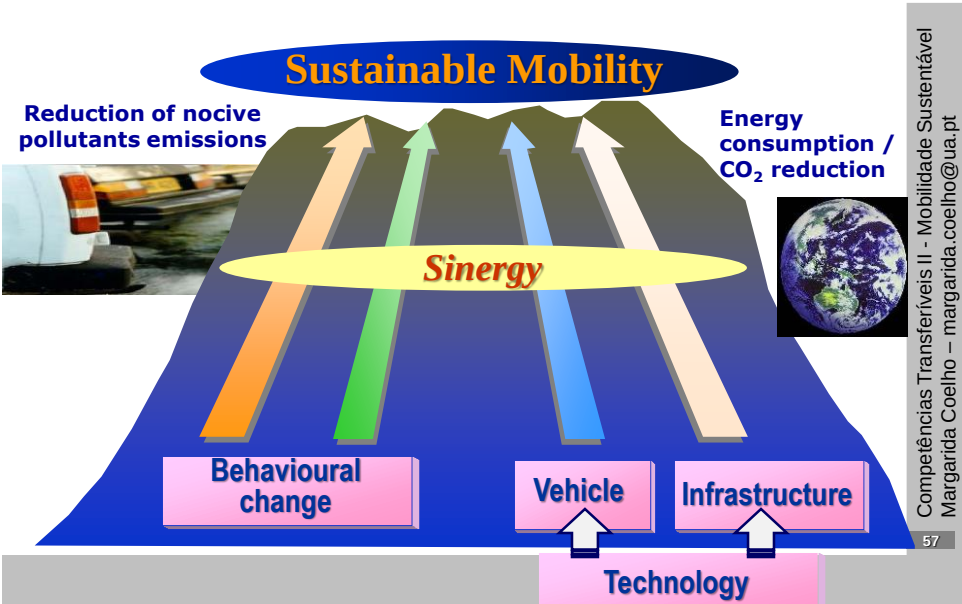


THE ENERGY ECOSYSTEM FOR TRANSPORT OF THE FUTURE





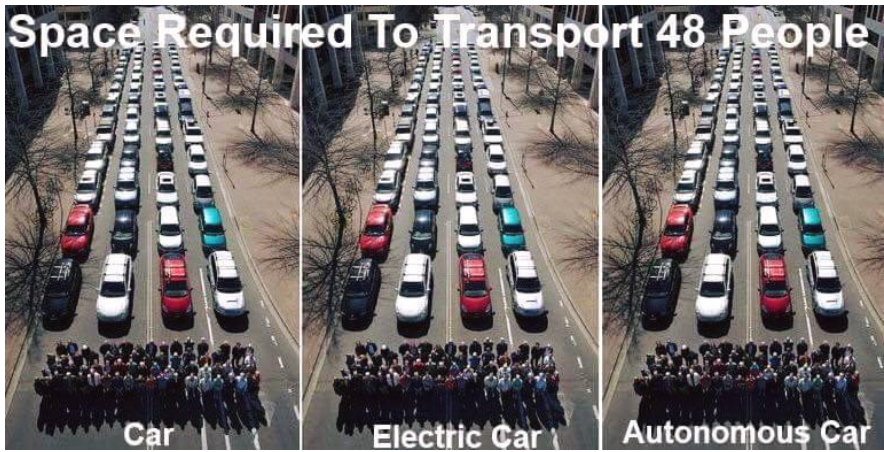
INTEGRATION OF ≠ MEASURES ⇒ A STRATEGICAL CHOICE
FOR SUSTAINABLE MOBILITY



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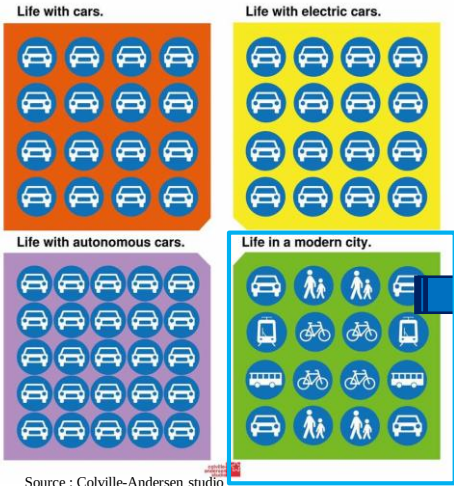
TECHNOLOGY DOES NOT CHANGE
GEOMETRY!



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WHAT IS THE MOBILITY THAT WE REALLY WANT FOR OUR FUTURE?

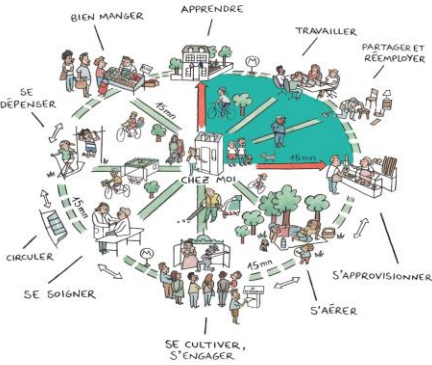


Source : Colville-Andersen studio

Smart Mobility → *inclusive, equitable, safe and sustainable*

tentável

LE PARIS DU 1/4 HEURE



Source : Citylab.com